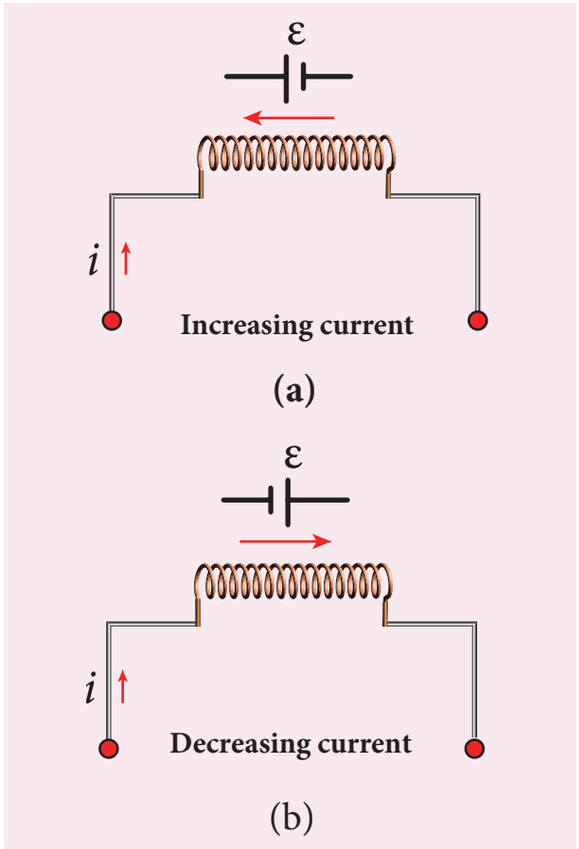


Figure 4.20 Induced emf ε opposes the changing current i

4.3.2 Self-inductance of a long solenoid

Consider a long solenoid of length l and cross-sectional area A. Let n be the number of turns per unit length (or turn density) of the solenoid. When an electric current i is passed through the solenoid, a magnetic field is produced by it which is almost uniform and is directed along the axis of the solenold as shown in Figure 4.21. The magnetic field at any point inside the solenoid is given by (Refer section 3.9.3)

$$B = \mu_{\circ} ni$$

As this magnetic field passes through the solenoid, the windings of the solenoid are linked by the field lines. The magnetic flux passing through each turn is

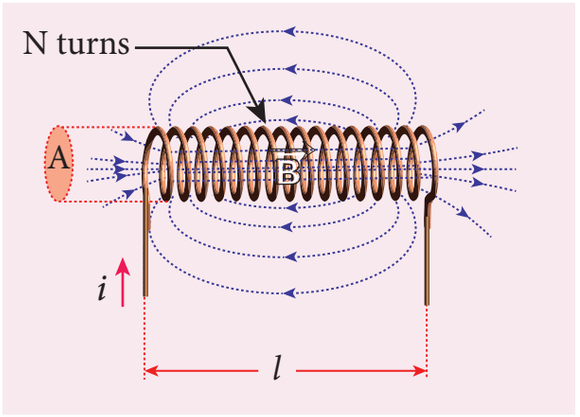


Figure 4.21 Self-inductance of a long solenoid

$$\Phi_B = \int_A \vec{B}.d\vec{A} = BA\cos\theta = BA \text{ since } \theta = 0°$$
$$= (\mu_0 ni)A$$

The total magnetic flux linked or flux linkage of the solenoid with N turns (the total number of turns N is given by $N = n\,l$) is

$$N\Phi_B = (nl)(\mu_0 ni)\,\mathrm{A}$$
$$N\Phi_B = (\mu_{\circ} n^2 Al)\, i \qquad (4.17)$$

The equation (4.15) is

$$N\Phi_B = Li$$

Comparing equations (4.15) and (4.17), we have

$$L = \mu_{\circ} n^2 Al$$

From the above equation, it is clear that inductance depends on the geometry of the solenoid (turn density n, cross-sectional area A, length l) and the medium present inside the solenoid. If the solenoid is filled with a dielectric medium of relative permeability μ_r, then

$$L = \mu n^2 Al \text{ or } \quad L = \mu_{\circ}\mu_r n^2 Al$$

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